



# AI-Based Grocery Expiry Alert System

An intelligent real-time application using AI logic to prevent the sale of expired items, safeguard health, and minimize food waste via barcode technology.

[ai-grocery-expiry-alert.vercel.app](https://ai-grocery-expiry-alert.vercel.app)

# The Problem Statement



Addressing the critical gap in retail food safety and environmental sustainability.

# THE PROBLEM WE SOLVE

- ⚠ **Undetected Expired Products:** Consumers often purchase risky items without checking dates manually.
- 🕒 **Manual Tracking Failures:** Staff reliance on visual shelf checks leads to frequent human errors.
- 🚫 **Zero POS Control:** Standard billing systems allow expired goods to pass through checkout silently.



# Our AI-Powered Solution



Automating food safety through intelligent machine-driven logic.

# | 3-TIER SMART DECISION SYSTEM

## Real-Time Intelligence

Our core AI engine evaluates product data against the current date in milliseconds, providing an unbreakable safety net at the point of sale.

**Automated · Fast · Accurate**



# CORE WEB APPLICATION FEATURES



## Camera Scanner

Instant barcode lookup using browser-native camera access on any device.



## AI Logic Engine

Automated decision logic blocking expired items and alerting for "near-expiry".



## Smart Billing

Dynamic cart that filters out unsafe items and generates digital invoices.



## Admin Control

Comprehensive dashboard to manage price, inventory, and expiry records.



## Live Dashboard

Real-time analytics showing store health and critical expiration alerts.

## Responsive UI

Fully browser-based, hardware-agnostic, and ready for mobile or desktop.

# | SEAMLESS BARCODE ENTRY

## The Entry Point

Our scanner supports both live camera feeds and manual text input to ensure operational continuity in all hardware environments.

- ⚡ Instant AI evaluation upon scan
- 🎮 Color-coded status feedback
- 🛒 Automatic cart integration



# ADMIN INVENTORY CONTROL HUB

| Barcode    | Product Name    | Price   | Expiry Date | AI Status |
|------------|-----------------|---------|-------------|-----------|
| 8901234... | Amul Milk 1L    | ₹62.00  | 22 Apr 2026 | VALID     |
| 8901111... | Sourdough Bread | ₹45.00  | 21 Apr 2026 | EXPIRING  |
| 8909876... | Organic Curd    | ₹38.00  | 19 Apr 2026 | EXPIRED   |
| 8902222... | Basmati Rice    | ₹160.00 | 30 Jun 2026 | VALID     |
| 8903333... | Salted Butter   | ₹55.00  | 15 May 2026 | VALID     |

# | LIVE EXPIRY INTELLIGENCE

24

TOTAL SKUS

18

SAFE STOCK

4

NEAR EXPIRY

2

EXPIRED (BLOCKED)

*Managers get a bird's-eye view of store health, updated in real-time as items are scanned.*

# 3-TIER AI LOGIC BREAKDOWN



## EXPIRED

Date < Today

**Action:** Billing completely blocked. Alert issued.  
Product must be discarded.



## EXPIRING SOON

Date  $\leq$  3 Days

**Action:** Warning issued to cashier. Item allowed with caution indicator.

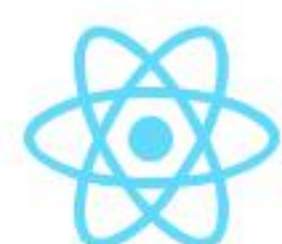


## VALID

Date > 3 Days

**Action:** Normal billing allowed. Seamless checkout process.

# MODERN TECHNOLOGY STACK



## Next.js

Server-side rendering for lightning-fast transitions and SEO.



## Vercel

Edge deployment ensuring global availability and low latency.



## Firebase

Real-time database powering instant inventory updates.

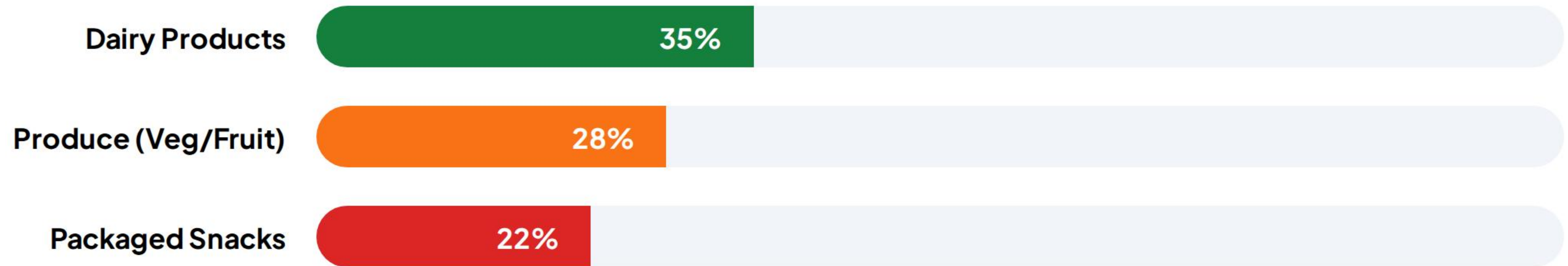
+ **Camera API** for scanning and **Tailwind CSS** for responsive design.

# MARKET READINESS & DEMAND



Data from a diverse sample of 100 respondents shows overwhelming tech receptivity.

# THE COST OF EXPIRY WASTE



High-frequency shopping habits drive the need for real-time digital tracking.

# | KEY SYSTEM BENEFITS

## Impact Beyond Profit

Our system doesn't just block sales; it educates consumers and empowers retailers to operate sustainably.

 **Environmental:** Significant reduction in landfill waste.

 **Economic:** Retailers avoid liability and stock loss.

 **Health:** Zero risk of consuming expired goods.

 Clean grocery store shelving

# The Future of Retail

Sustainable grocery management isn't just a goal—it's a responsibility. AI-Based Expiry Alerts pave the way for a waste-free future.



# Questions?

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Thank you for your time and attention.

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Smart · Safe · Sustainable

# IMAGE SOURCES



<https://www.thetakeout.com/img/gallery/stop-throwing-these-12-foods-out-they-dont-expire-as-fast-as-you-think/intro-1762597769.jpg>

Source: [www.thetakeout.com](http://www.thetakeout.com)

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[https://lifehacker.com/imagery/articles/01KDNHTX3BKPDR6QF6FCMOD8D/hero-image.fill.size\\_1200x675.jpg](https://lifehacker.com/imagery/articles/01KDNHTX3BKPDR6QF6FCMOD8D/hero-image.fill.size_1200x675.jpg)

Source: [lifehacker.com](http://lifehacker.com)

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[https://png.pngtree.com/thumb\\_back/fh260/background/20251209/pngtree-close-up-of-a-hand-holding-barcode-scanner-with-red-laser-image\\_20782536.webp](https://png.pngtree.com/thumb_back/fh260/background/20251209/pngtree-close-up-of-a-hand-holding-barcode-scanner-with-red-laser-image_20782536.webp)

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[https://images.stockcake.com/public/8/4/b/84bc13ba-993f-48da-89a2-36470dea352c\\_large/modern-grocery-experience-stockcake.jpg](https://images.stockcake.com/public/8/4/b/84bc13ba-993f-48da-89a2-36470dea352c_large/modern-grocery-experience-stockcake.jpg)

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